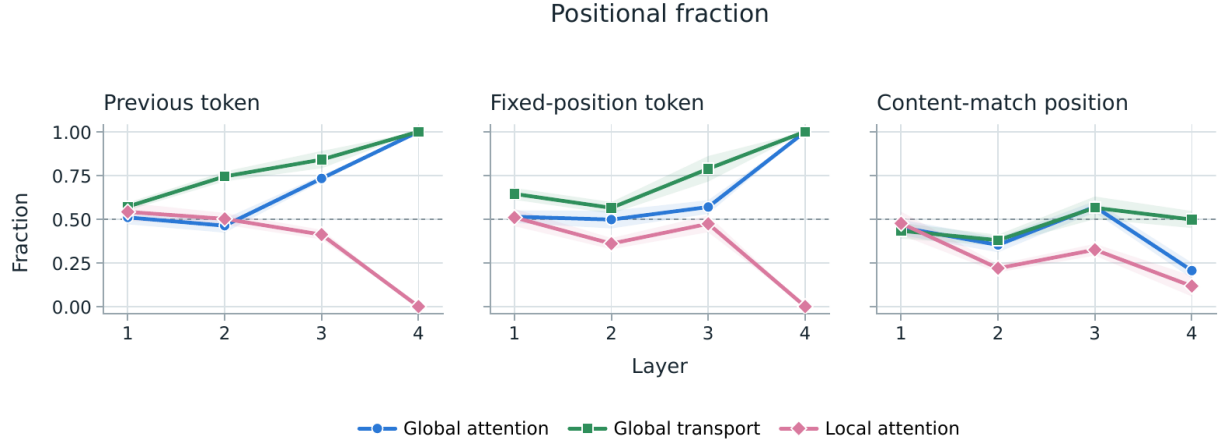
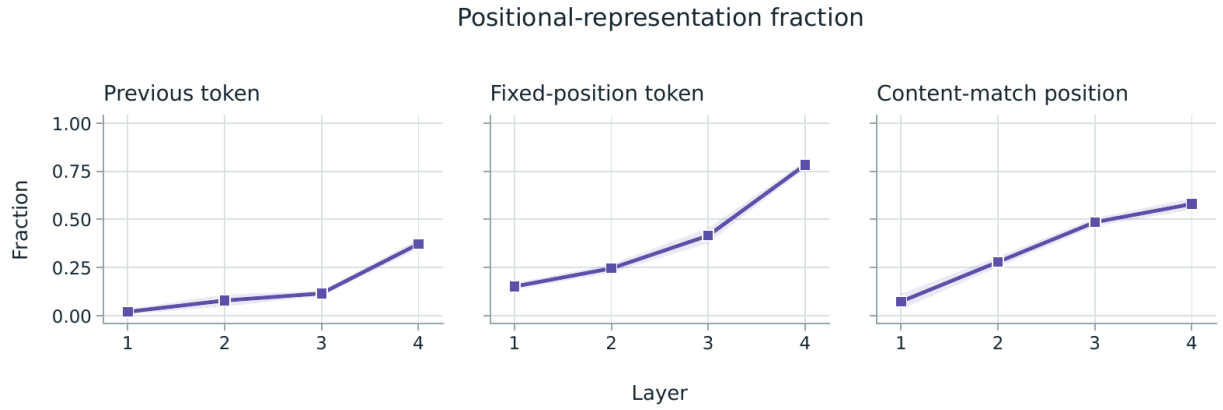


(a) Illustration of lookup operation tasks.



(b) Positional fraction for each scoring method and task.



(c) Residual positional-representation fraction.

Fig. 4.1 Synthetic lookup operation task.

(a) Illustrations of the teacher tasks. (b) Mean per-layer positional fraction, $p/(p+s)$, with shading showing ± 1 SD across six seeds. (c) Mean positional-representation fraction, with shading denoting ± 1 SD across six seeds.